Exp 2: Design and Optimization of Yagi-Uda Antenna using MMANA-GAL

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Aim: To design a 3-element Yagi-Uda antenna and optimize element lengths and spacing for maximum gain, minimum VSWR and improved front-to-back ratio.

Design Parameters

Element	Length (m)	X position (m)
Reflector	0.488	-0.168
Driven	0.461	0.000
Director	0.451	0.298

Geometry Table (MMANA-GAL)

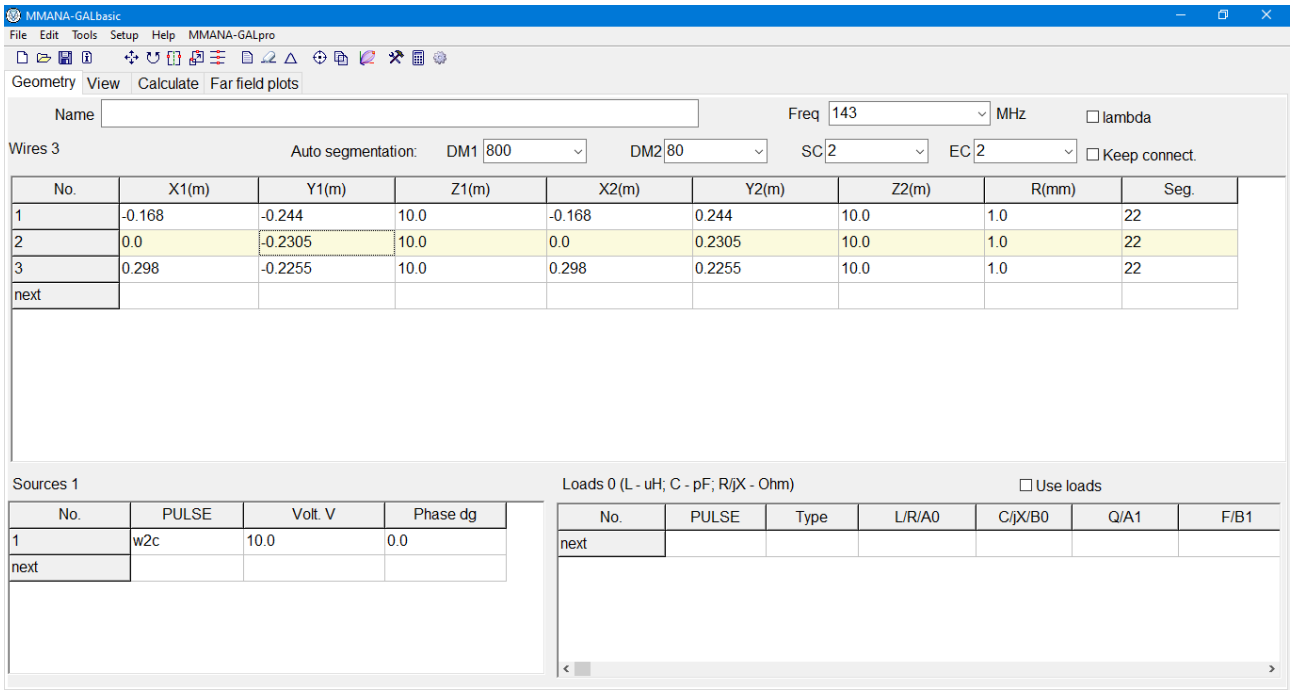


Fig. 1: Geometry tab of the Yagi-Uda antenna (3 wires, feed on wire 2 centre, 143 MHz)

Far Field Radiation Pattern

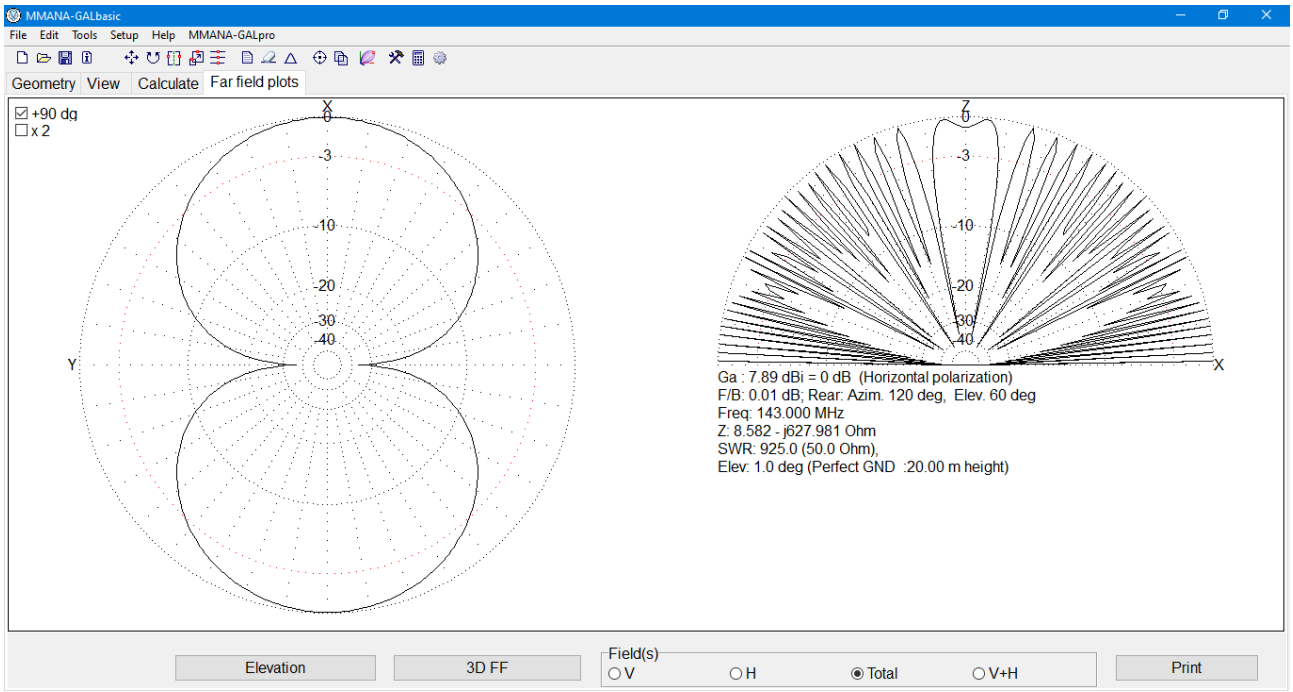


Fig. 2: Far field plot of the Yagi-Uda antenna. Left: azimuth. Right: elevation.